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Diagnosing feedback with the Lyman-alpha forest at $z \sim 0.1$

Jamie Bolton

With thanks to:

Fahad Nasir, Matteo Viel, Martin Haehnelt, Tae-Sun Kim, Ewald Puchwein,
Debora Sijacki



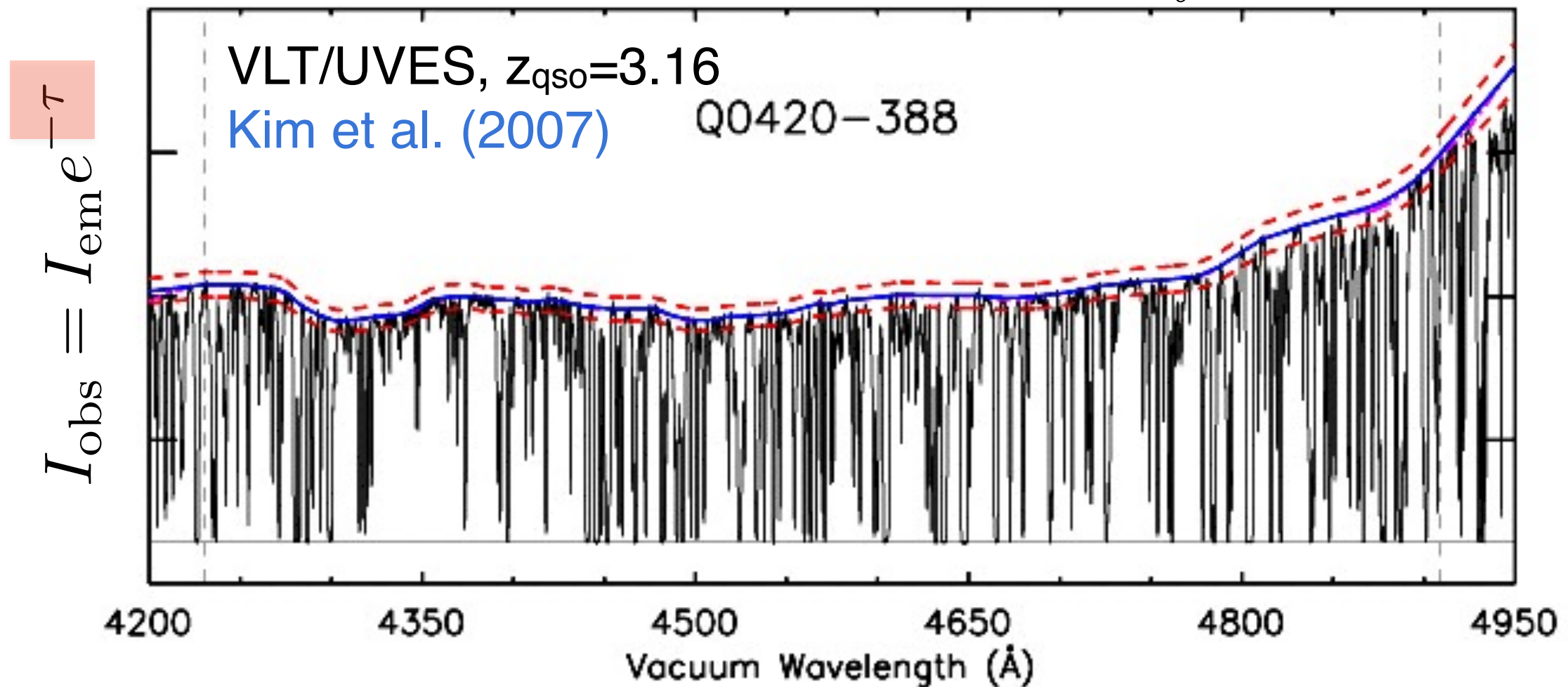
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The Lyman- α forest

$$\lambda_{\text{Ly}\alpha} = 1216 \text{ \AA}$$



Opacity

$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T_0^{0.7} H(z) \Gamma(z)} (1+\delta)^{2-0.7(\gamma-1)}$$

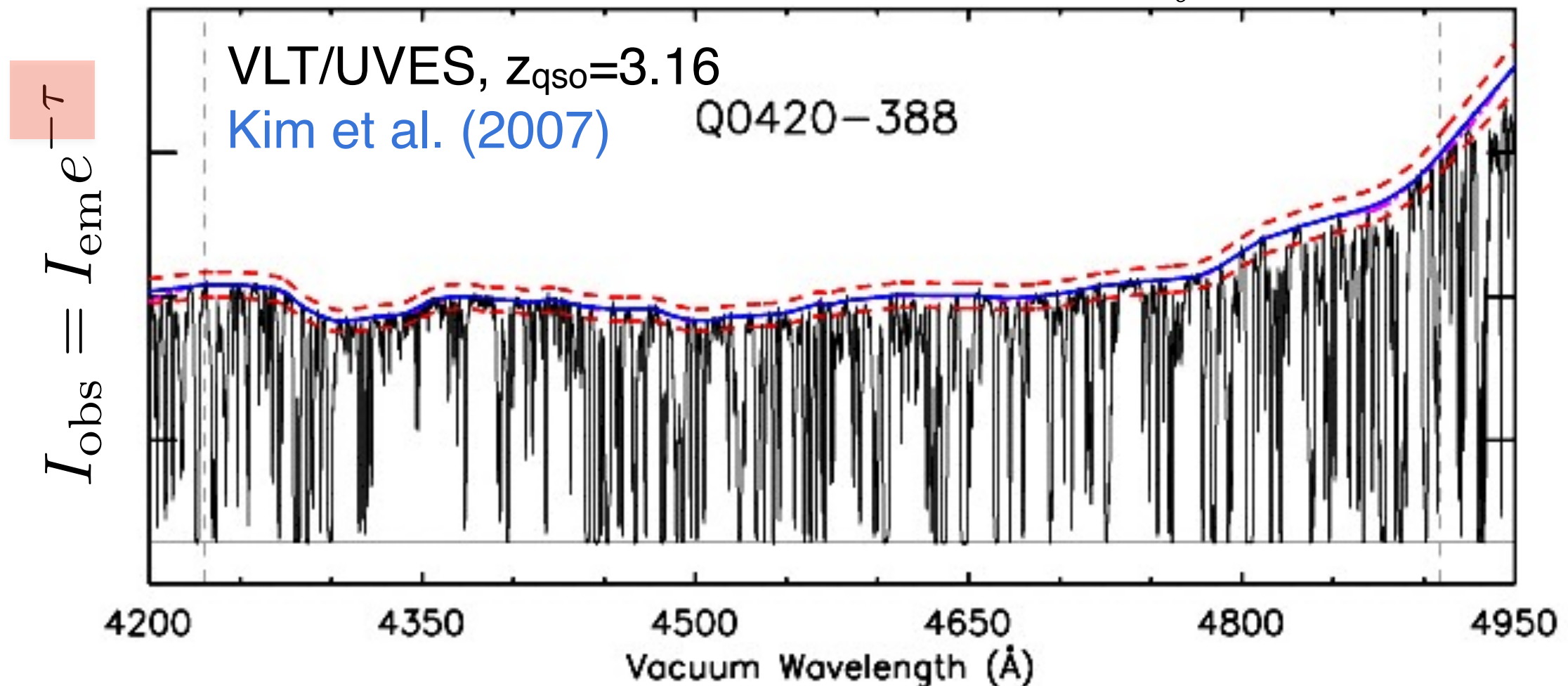
Temperature

Density

Hydrogen photo-ionisation rate

The Lyman- α forest

$$\lambda_{\text{Ly}\alpha} = 1216 \text{ \AA}$$



$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T_0^{0.7} H(z) \Gamma(z)} (1+\delta)^{2-0.7(\gamma-1)}$$

$\tau \sim 1$

$T \sim 10^4 \text{ K}$

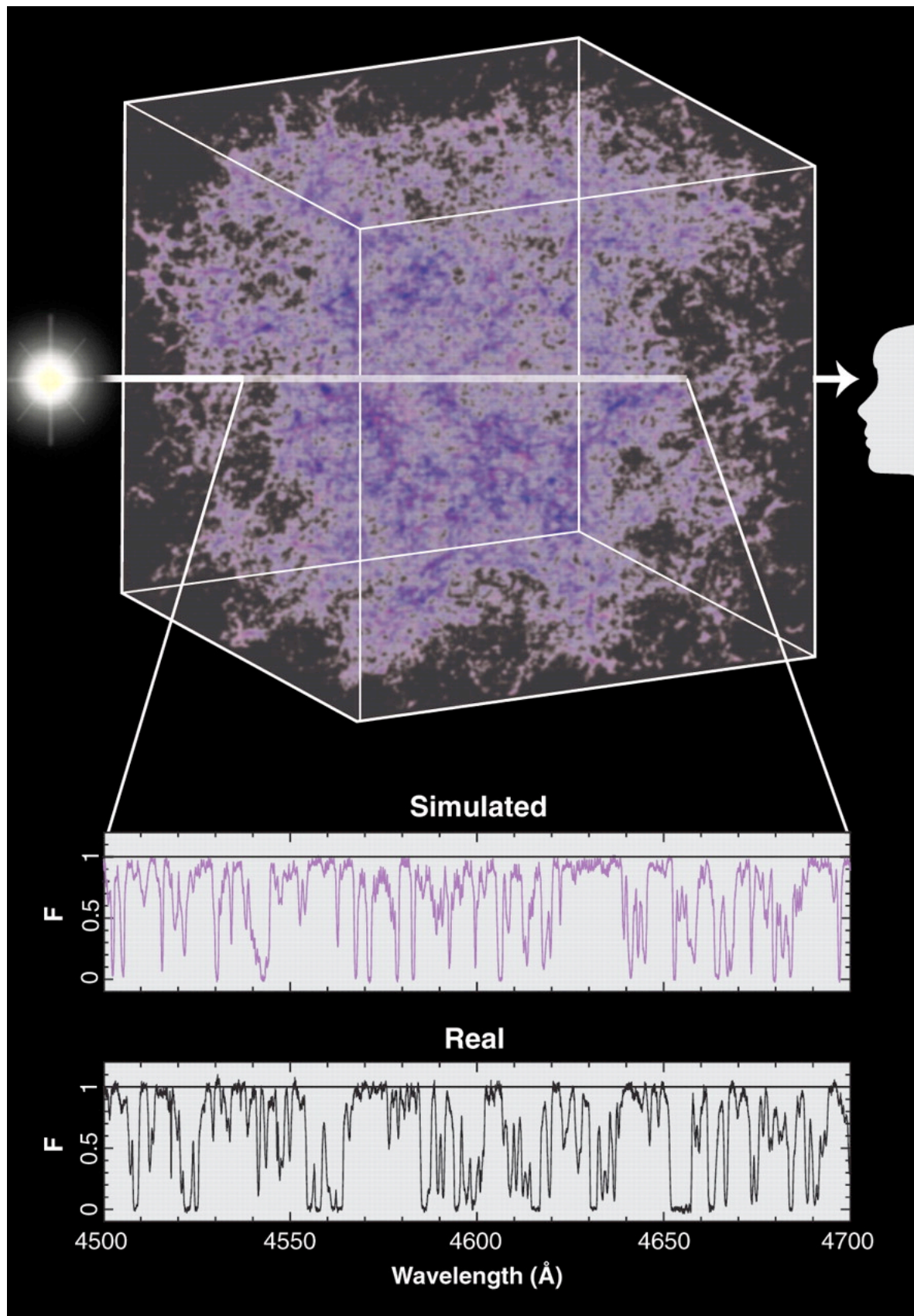
$\Gamma \sim 10^{-12} \text{ s}^{-1}$

$\delta \sim 1-10$

Minimal IGM recipe

- Cosmological parameters
- Matter power spectrum, $P(k)$
- Newtonian gravity
- Hydrodynamics (>100 kpc/h)
- Photo-ionisation and heating (hydrogen and helium only)
- Integrate w.r.t time (*lots...!*)

$\Rightarrow$ Lyman- α forest



Faucher-Giguere et al. (2008)



The Sherwood Simulation Suite

Hydrodynamical models of the
Lyman-alpha forest



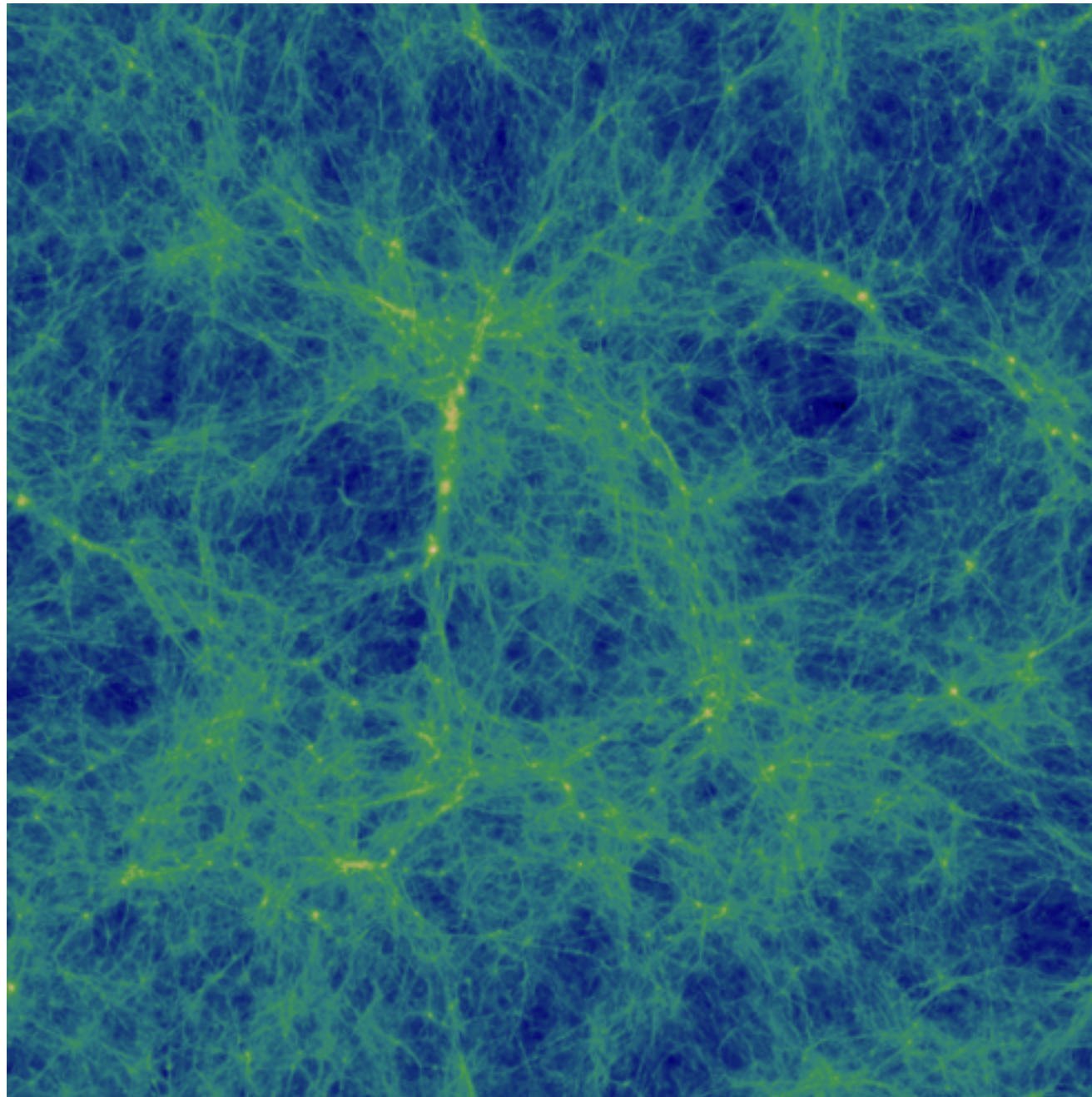
Science & Technology
Facilities Council



DiRAC

<http://www.nottingham.ac.uk/astronomy/sherwood/>

The Sherwood simulation suite

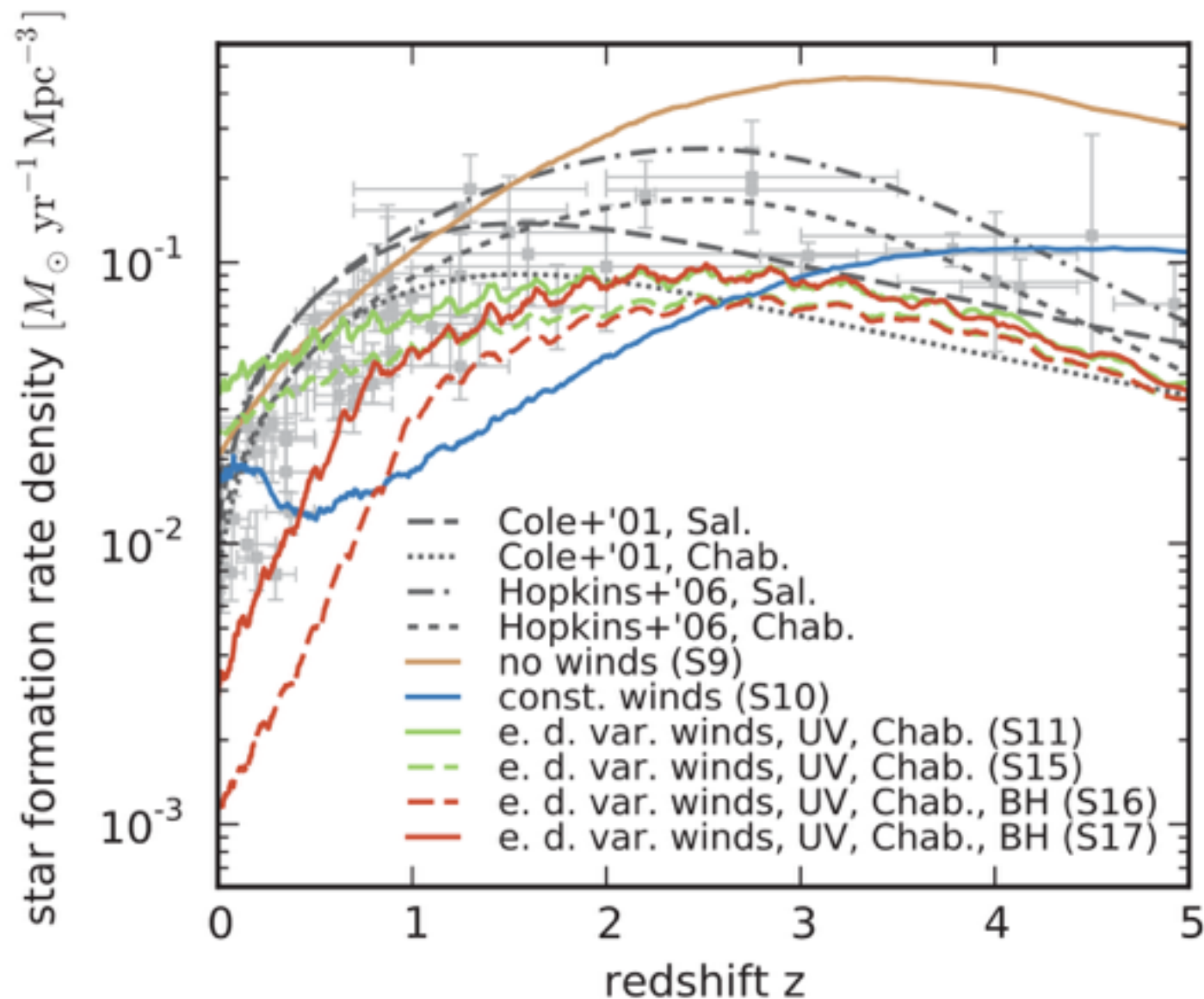


40 comoving Mpc/h at $z=2.8$

- 15 million hours on Curie through PRACE
- 40-160 Mpc/h boxes with 2×2048^3 particles; Planck cosmology; fully hydrodynamical IGM calculations with P-Gadget-3; ~ 70 Tb data
- CDM and WDM (3.5 keV thermal relic) runs, varying reionisation histories

See Bolton et al. (2017) for overview, and website for all publications (12 to date).

Feedback in Sherwood

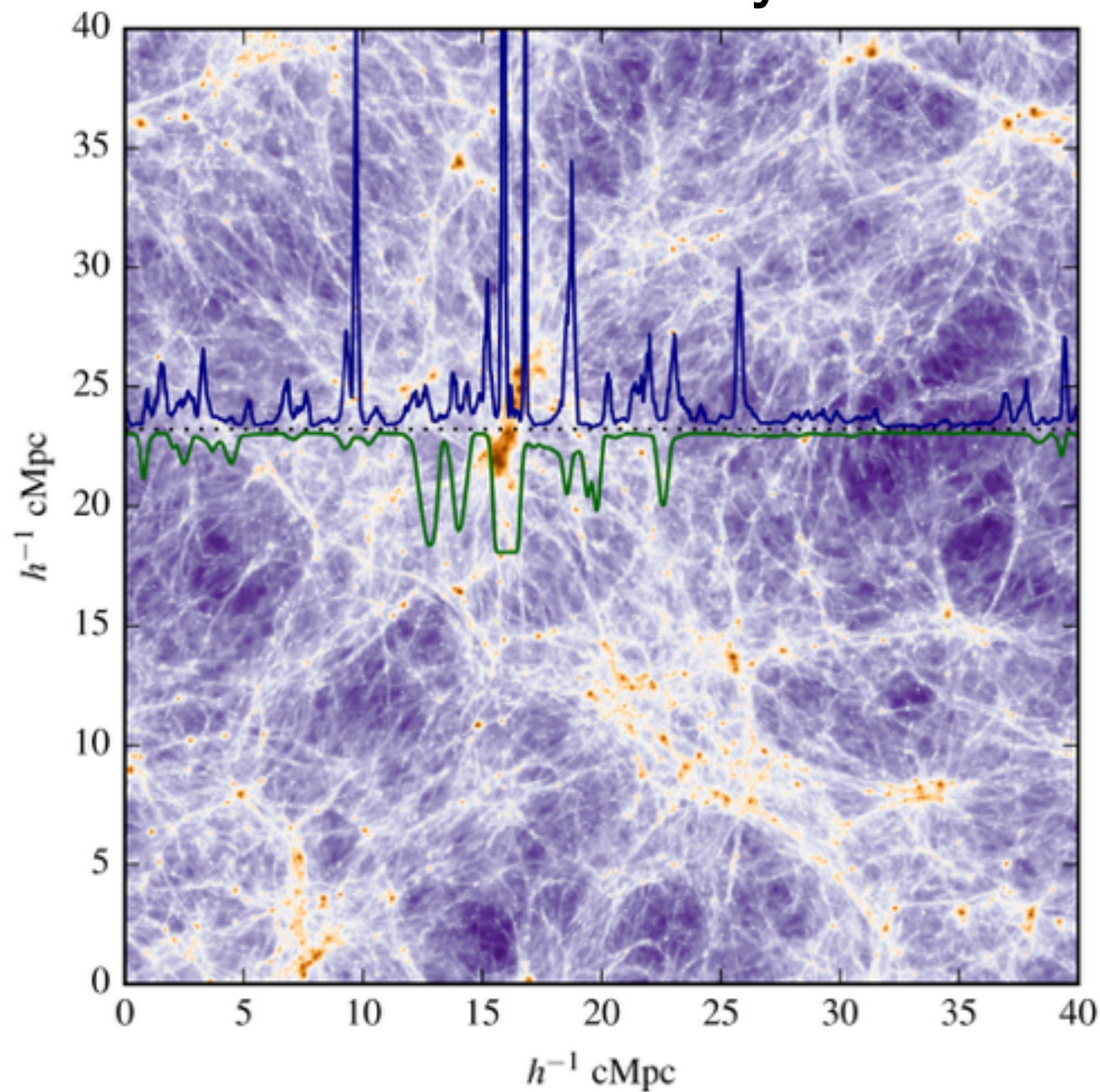


Puchwein & Springel (2013)

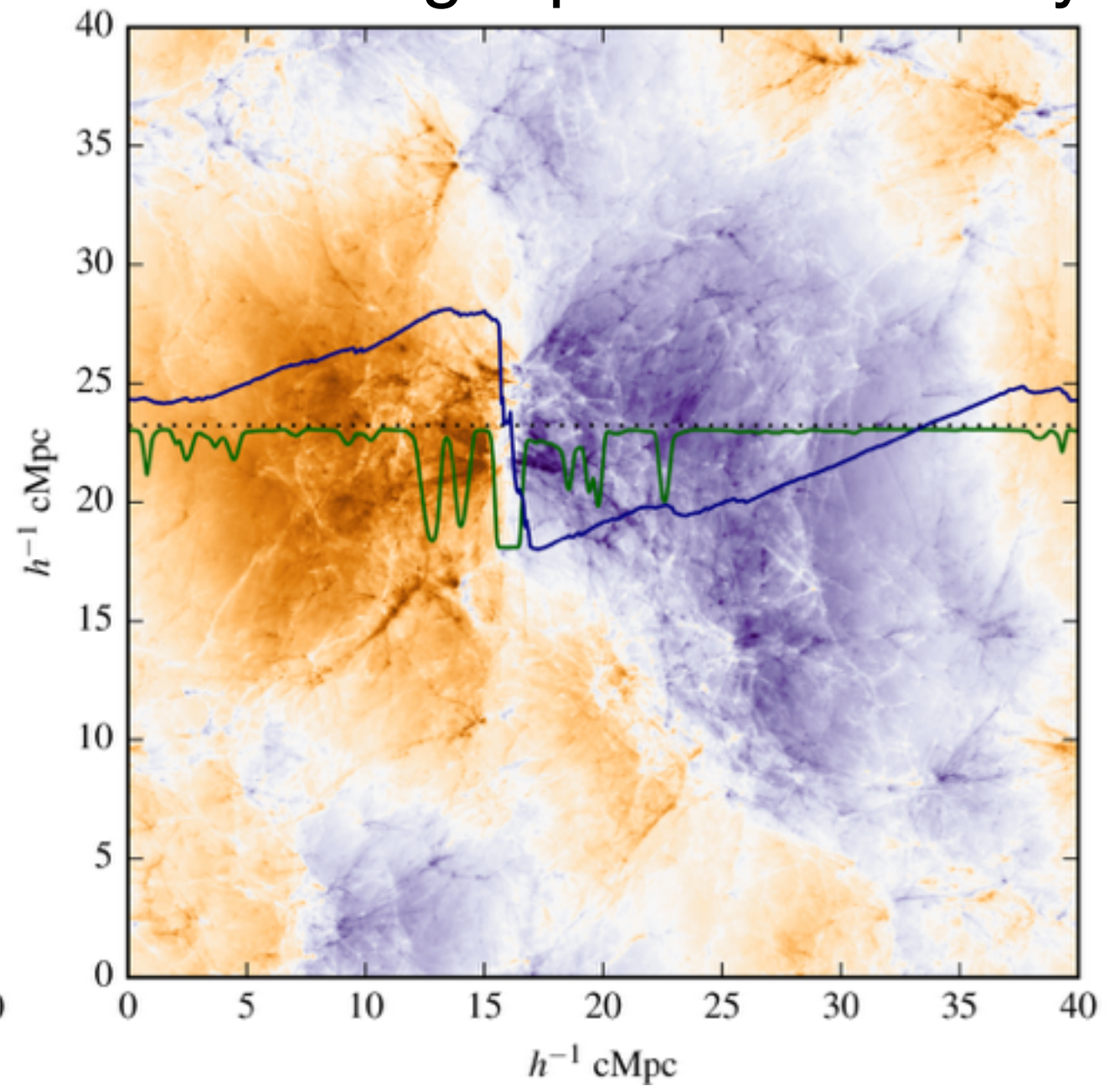
- Most Sherwood models adopt fast, approximate scheme that ignores star formation/feedback: “QUICKLYA” ([Viel et al. 2004](#));
- Stellar feedback, where included, follows [PS13](#) variable winds model; galactic wind velocity proportional to halo escape velocity;
- AGN feedback based on [Sijacki et al. \(2007\)](#);

Synthetic Lyman- α forest spectra

Gas density

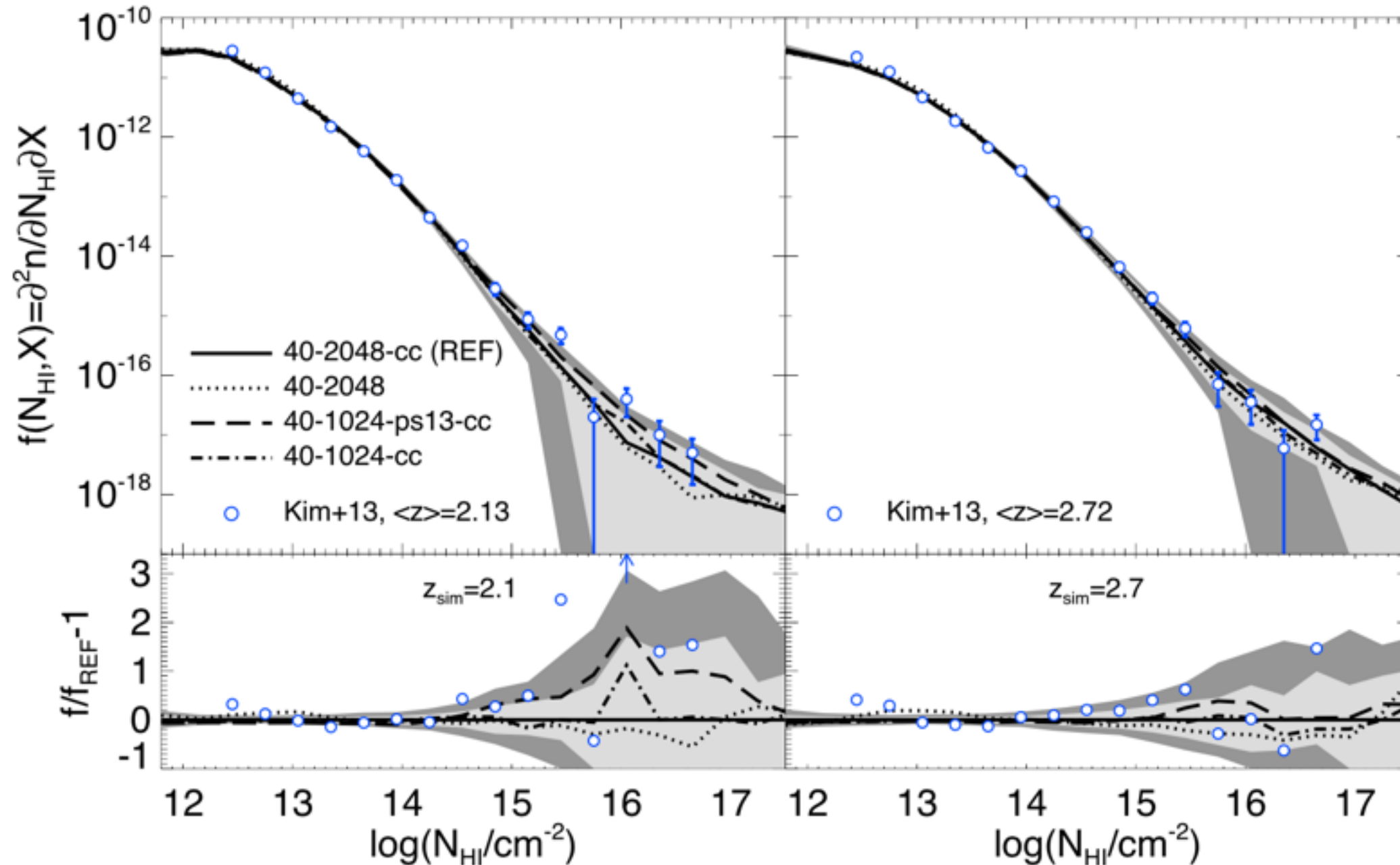


Line of sight peculiar velocity



Bolton et al. (2017)

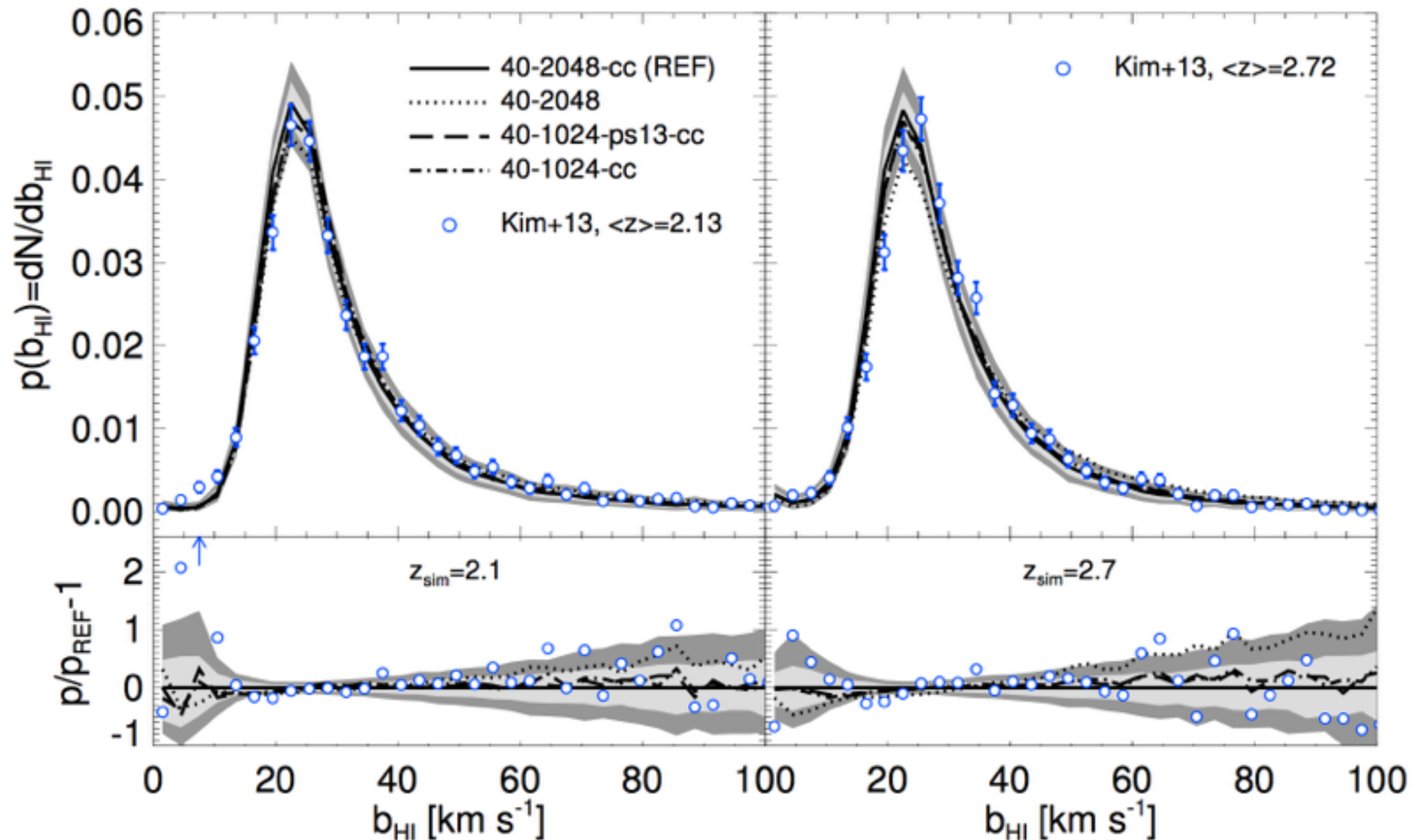
Column density distribution



Data from [Kim+13](#): 18 VLT/
UVES spectra with S/N~35-50

[Bolton et al. \(2017\)](#)

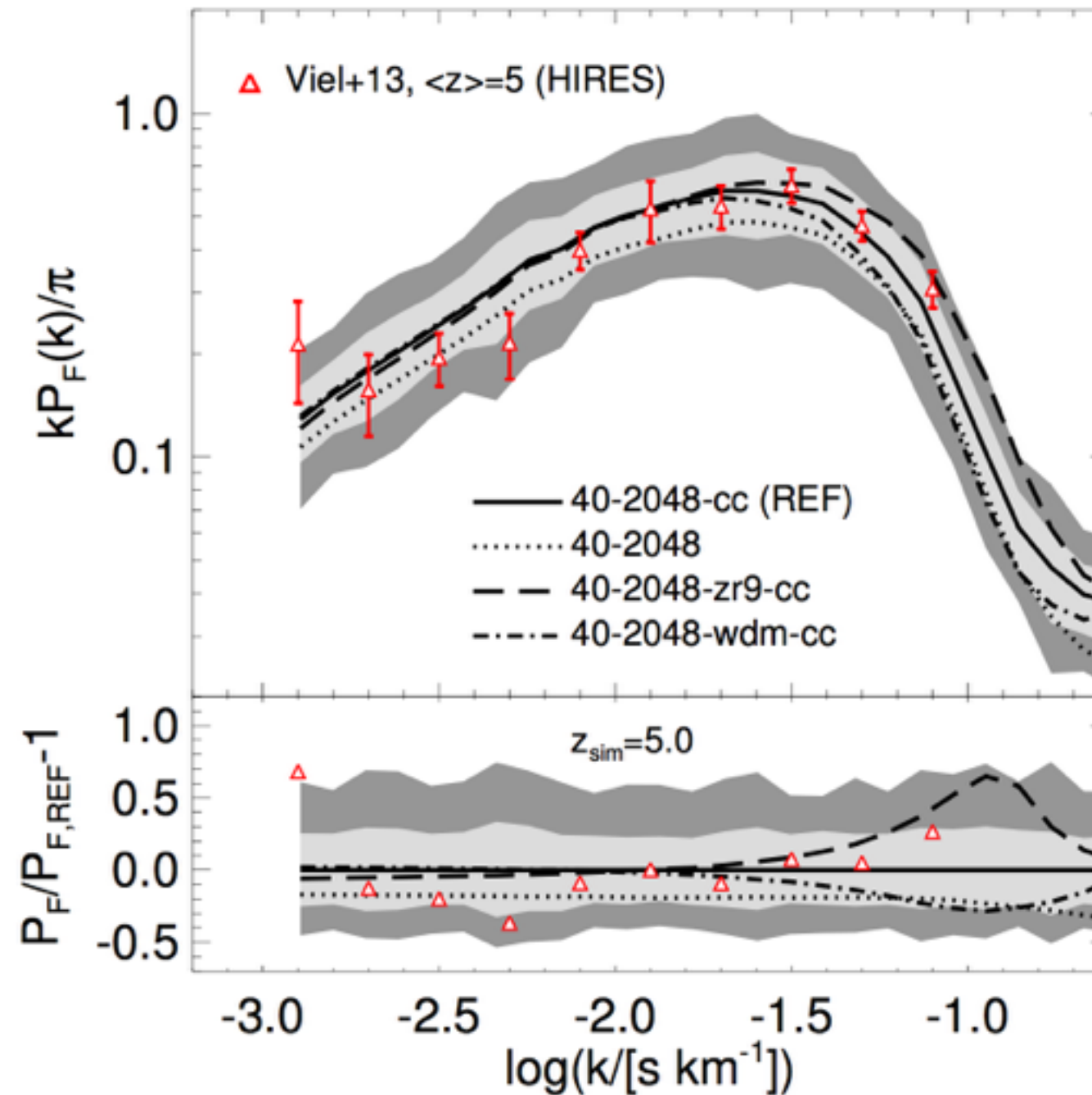
Line (Doppler) width distribution



Data from [Kim+13](#): 18 VLT/
UVES spectra with S/N~35-50

[Bolton et al. \(2017\)](#)

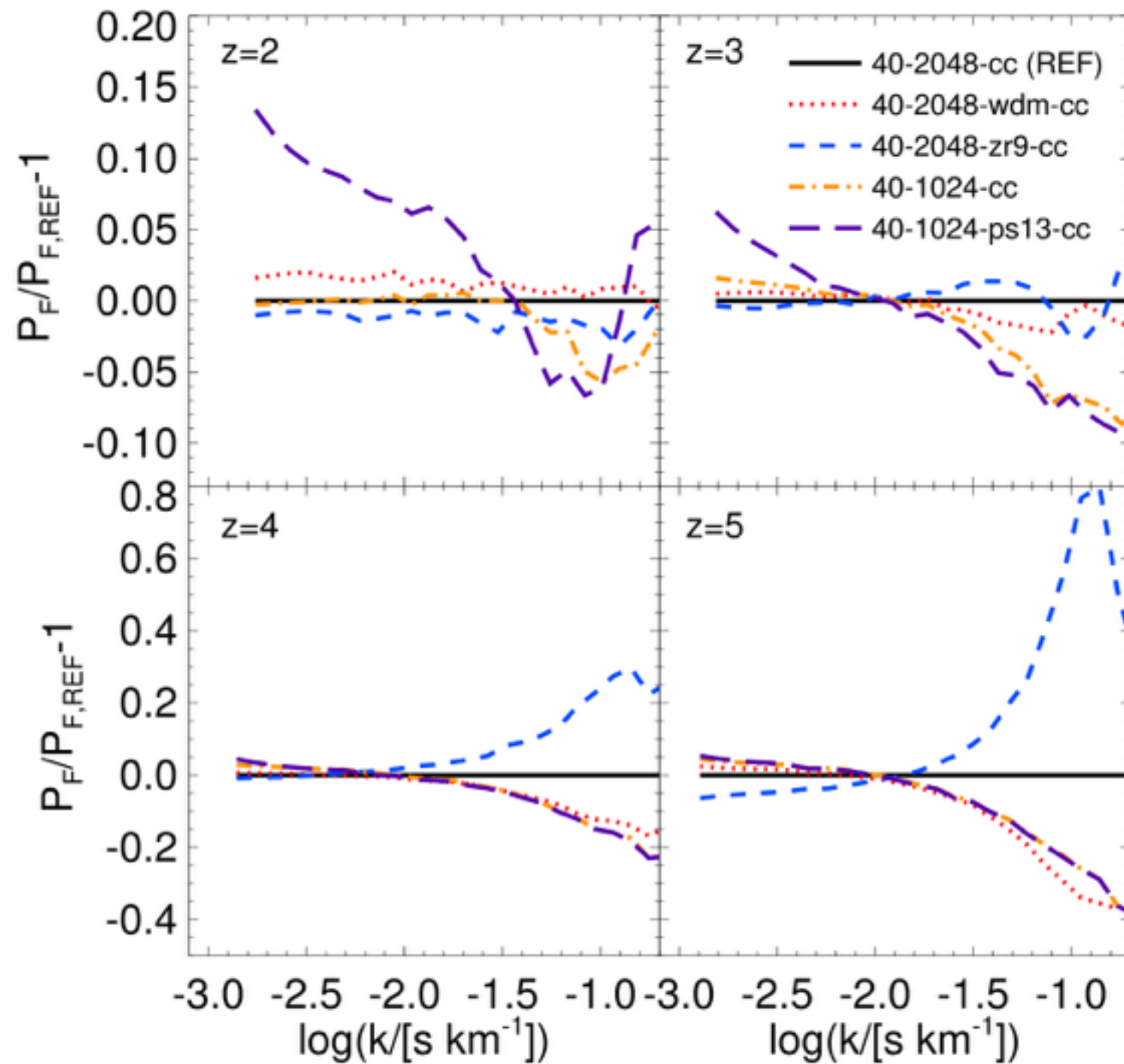
Transmission power spectrum



Data from [Viel+13](#): 14 Keck/
HIRES spectra with $S/N \sim 10-20$

[Bolton et al. \(2017\)](#)

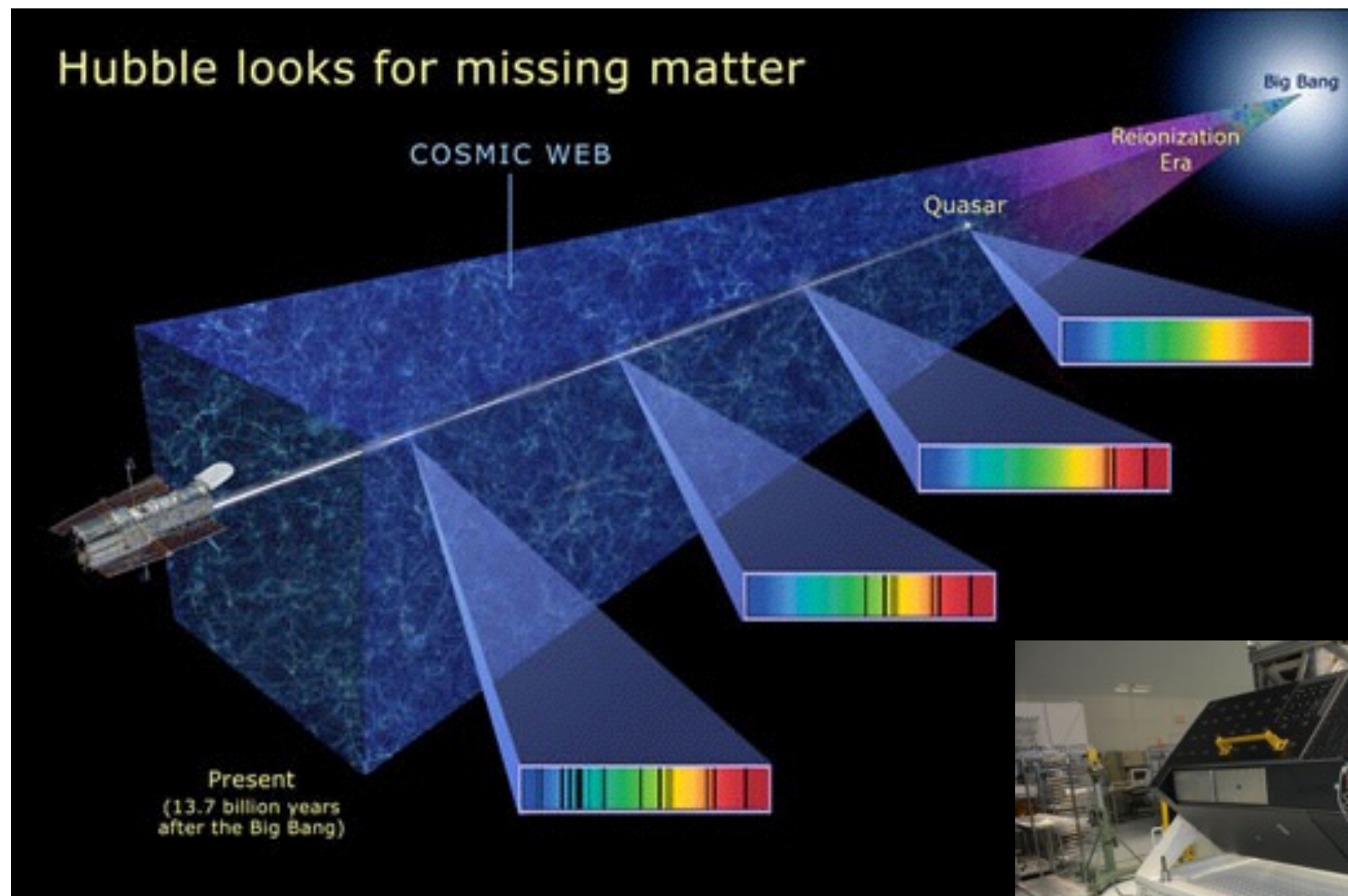
Transmission power spectrum



Summary, Part I

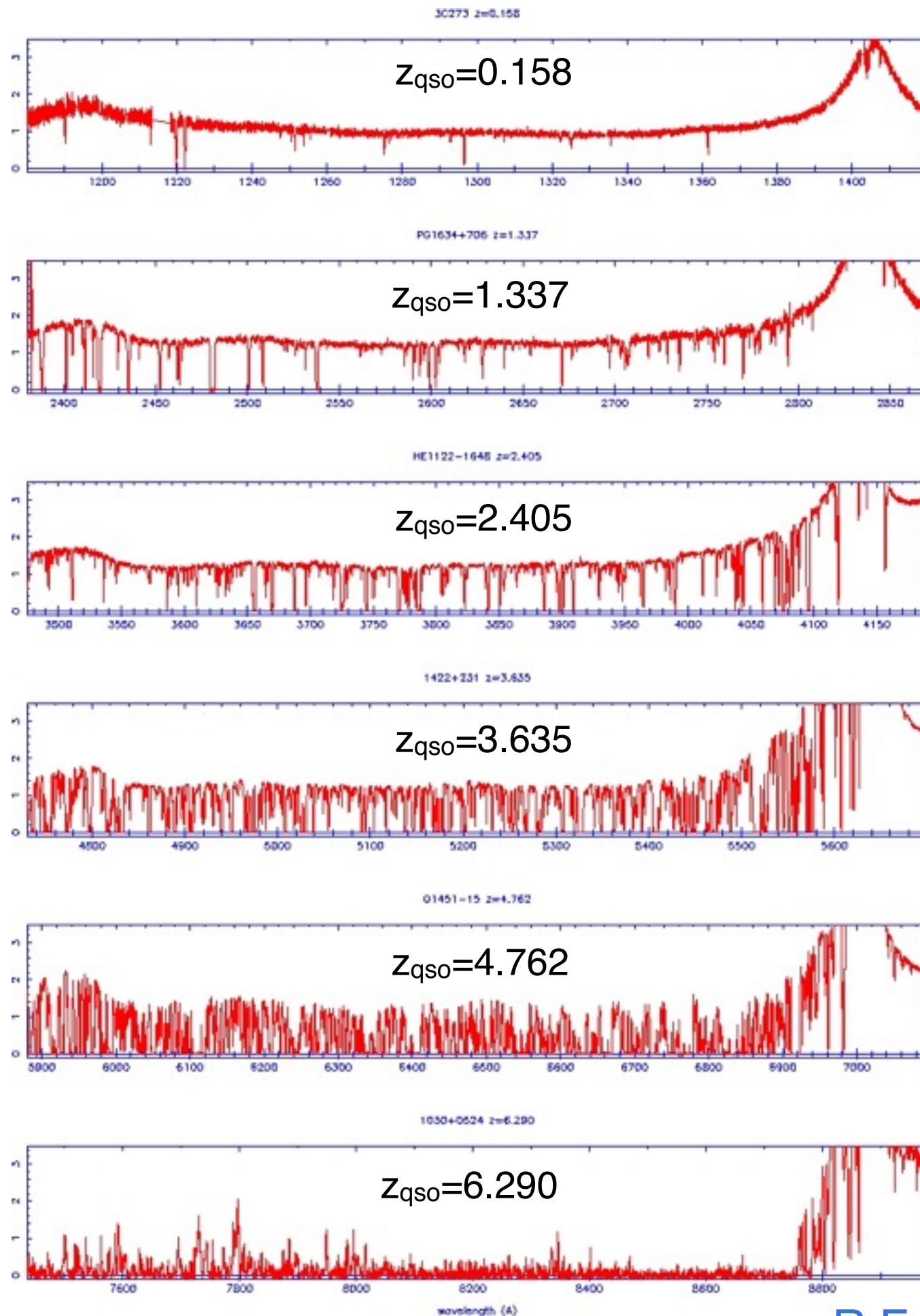
- Sherwood simulations reproduce a wide range of Lyman-alpha forest measurements very well at $2 < z < 5$;
- Star formation and galactic winds have a modest impact on saturated ($N_{\text{HI}} > 10^{14.5} \text{ cm}^{-2}$) lines/large scale power at $z < 2.5$ (see also [Theuns et al. 2002](#), [Viel et al. 2013a](#));
- Warm dark matter/thermal history significantly impacts on small-scale power at $z > 4$ ([Viel et al. 2013b](#), [Irsic et al. 2017](#)),

The Lyman- α forest at $z \sim 0.1$



$$\lambda_{\text{Ly}\alpha}(1+z) = 1338 \text{ \AA}$$

Need UV spectrograph (COS/HST)

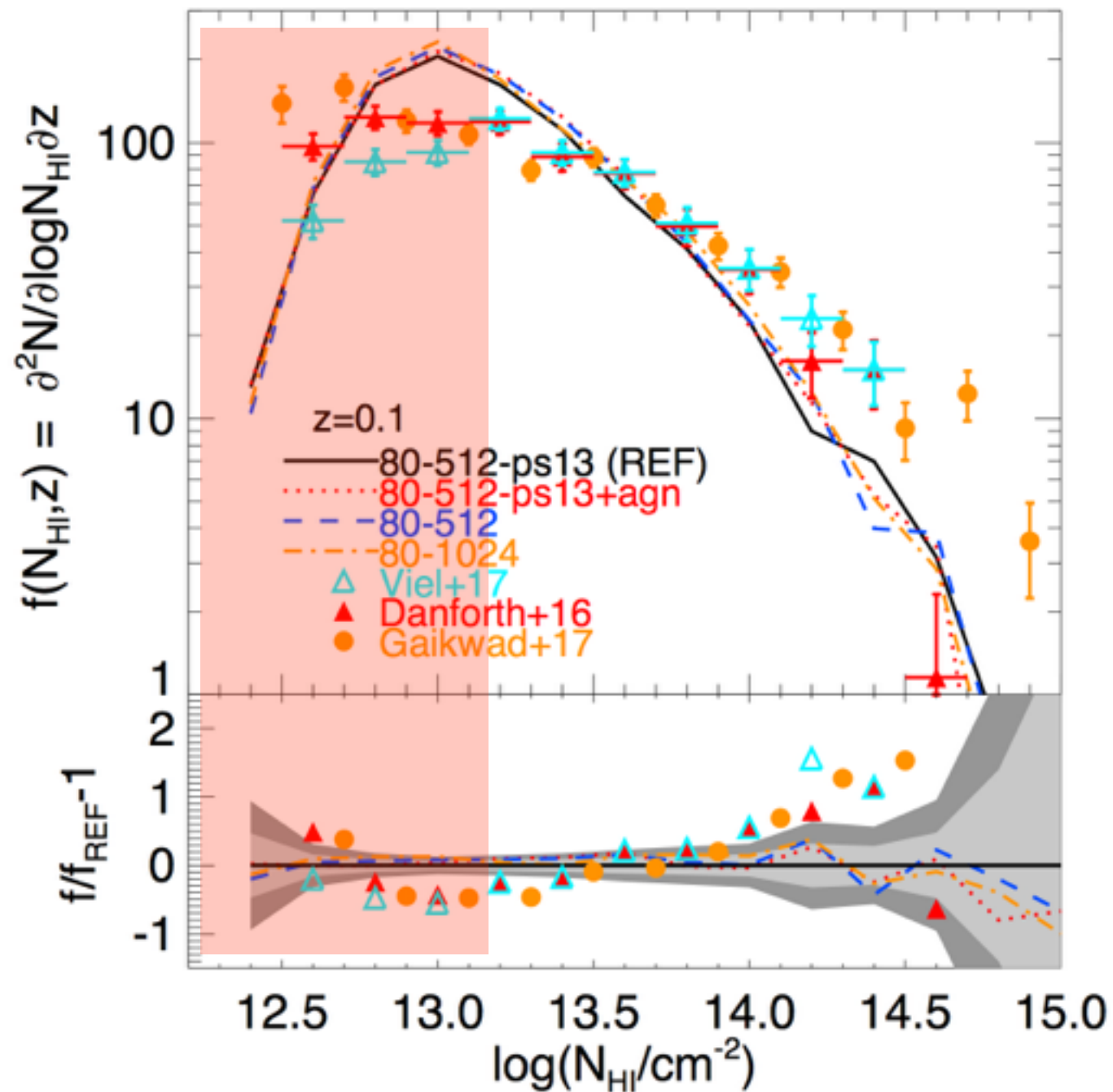
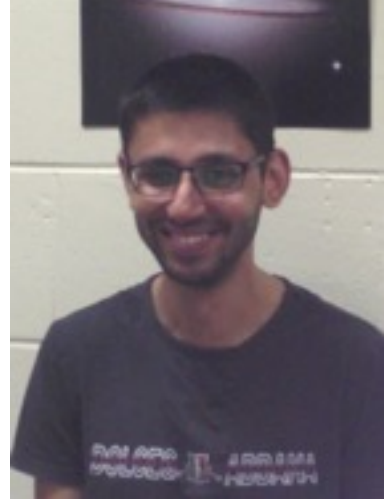


- Lyman-alpha forest thins significantly toward lower redshift, despite declining UV background amplitude;
- The proper HI density decreases;
- Structure formation draws more gas into dense regions, making voids larger and emptier;

$$\tau \propto (1 + z)^{9/2}$$

- Hence typical gas density probed by absorption increases.

Column density distribution



Nasir et al. (2017)

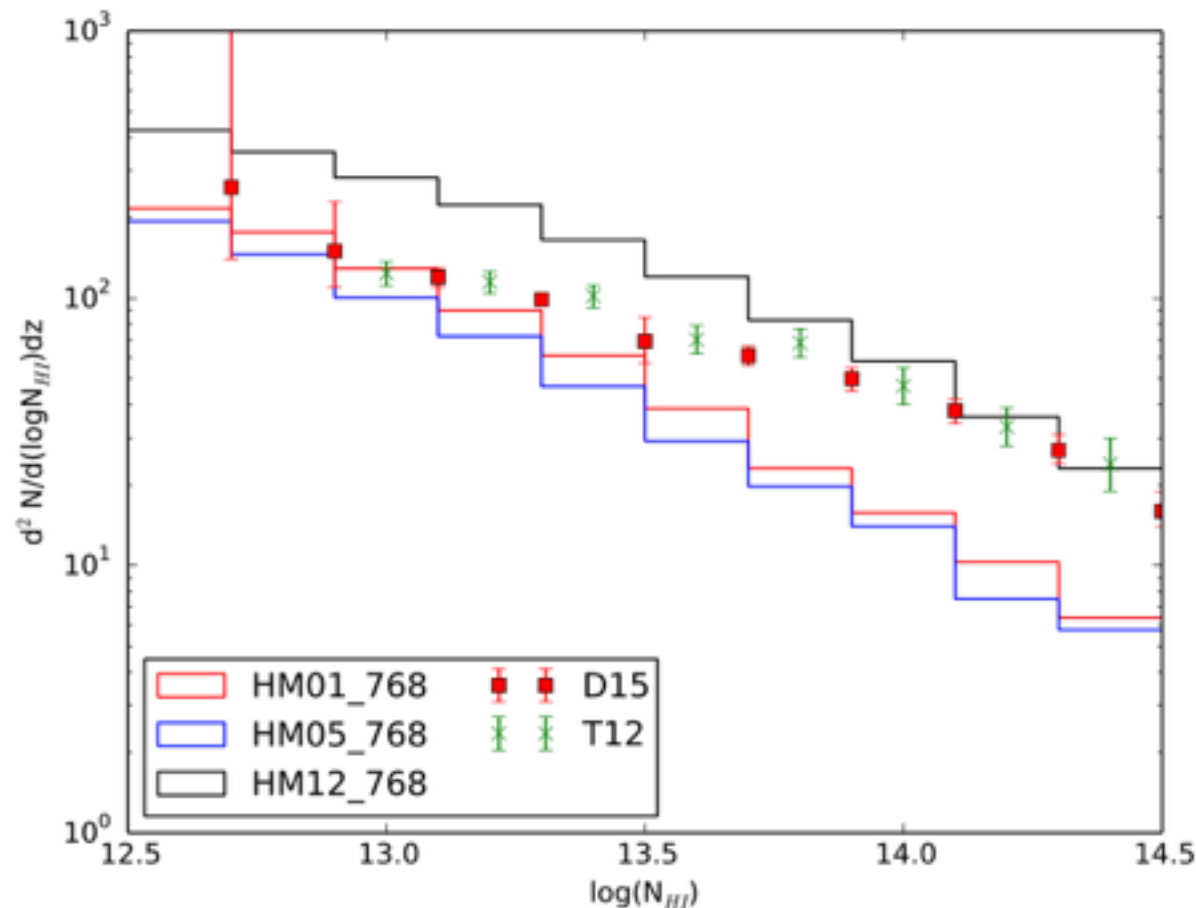
Viel+17 COS data: 704 lines
with $10^{12.5} < N_{\text{HI}} < 10^{14.5} \text{ cm}^{-2}$

- Feedback as implemented in Sherwood has very little impact on the CDDF;
- Simulations struggle to capture lines at $N_{\text{HI}} > 10^{14} \text{ cm}^{-2}$

A generic problem at $N_{\text{HI}} > 10^{14} \text{ cm}^{-2}$?

ENZO (AMR code)

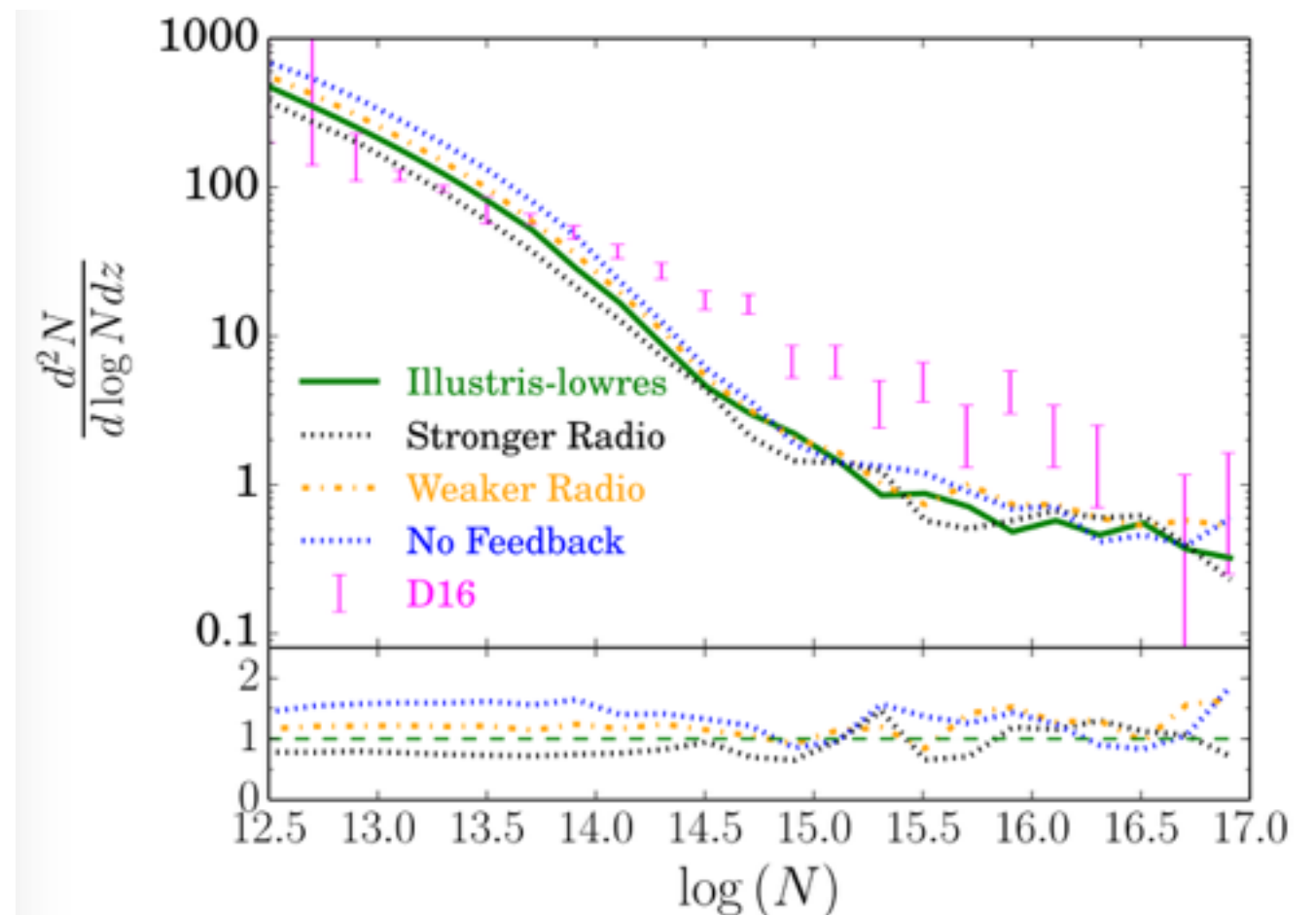
Varying UV backgrounds



Shull et al. (2015)

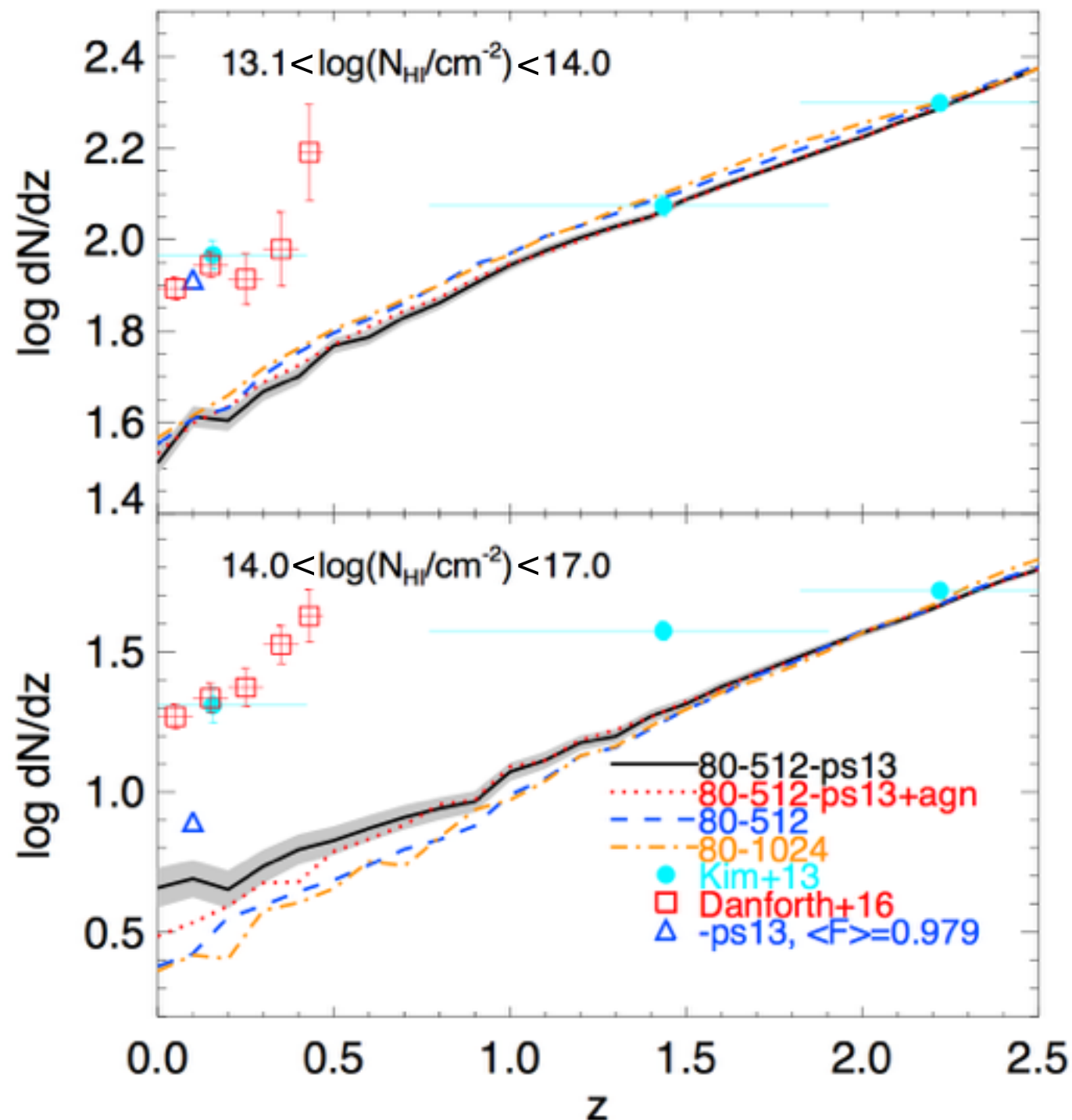
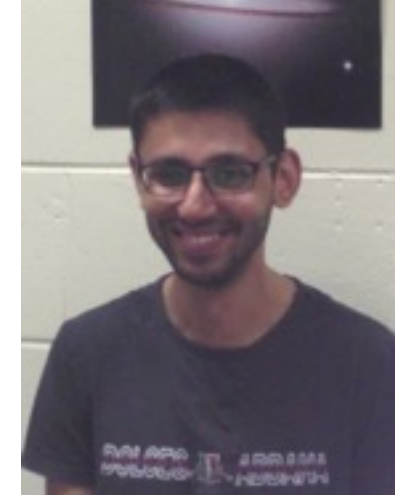
Arepo (moving mesh code)

Varying AGN feedback



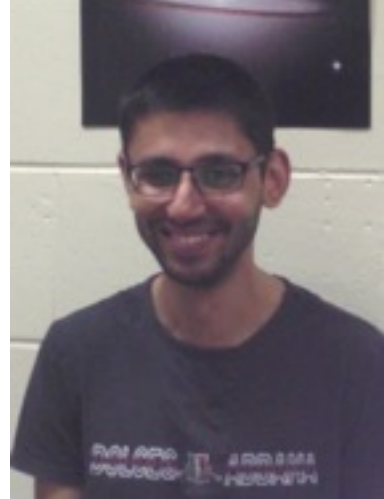
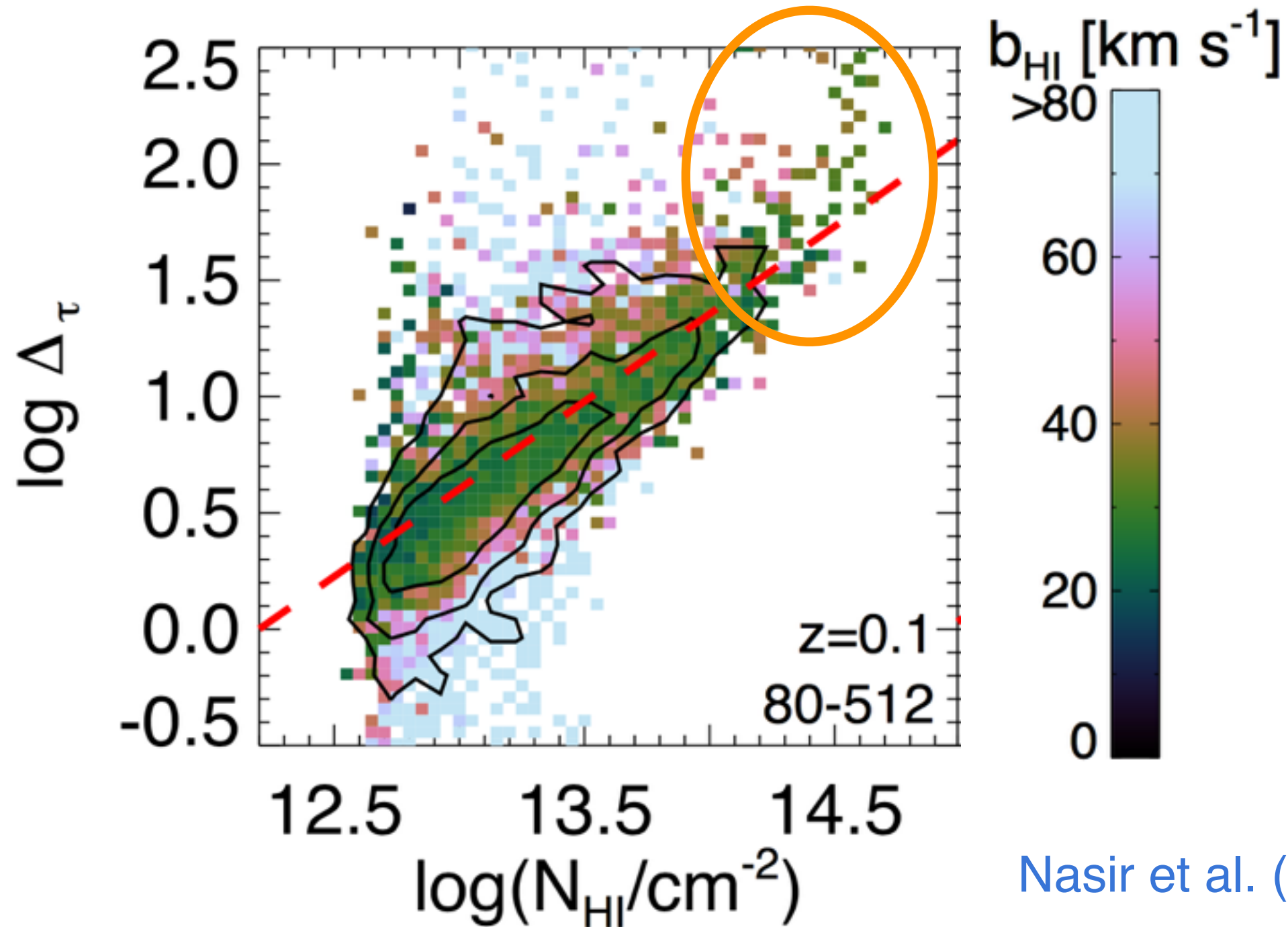
Gurvich et al. (2017)

Line number density



- Good agreement for dN/dz of unsaturated lines obtained on adjusting UV background amplitude;
- Stronger lines fall short by factor 2-3 at $z < 1.5$;
- Choice of feedback more important for strong lines.

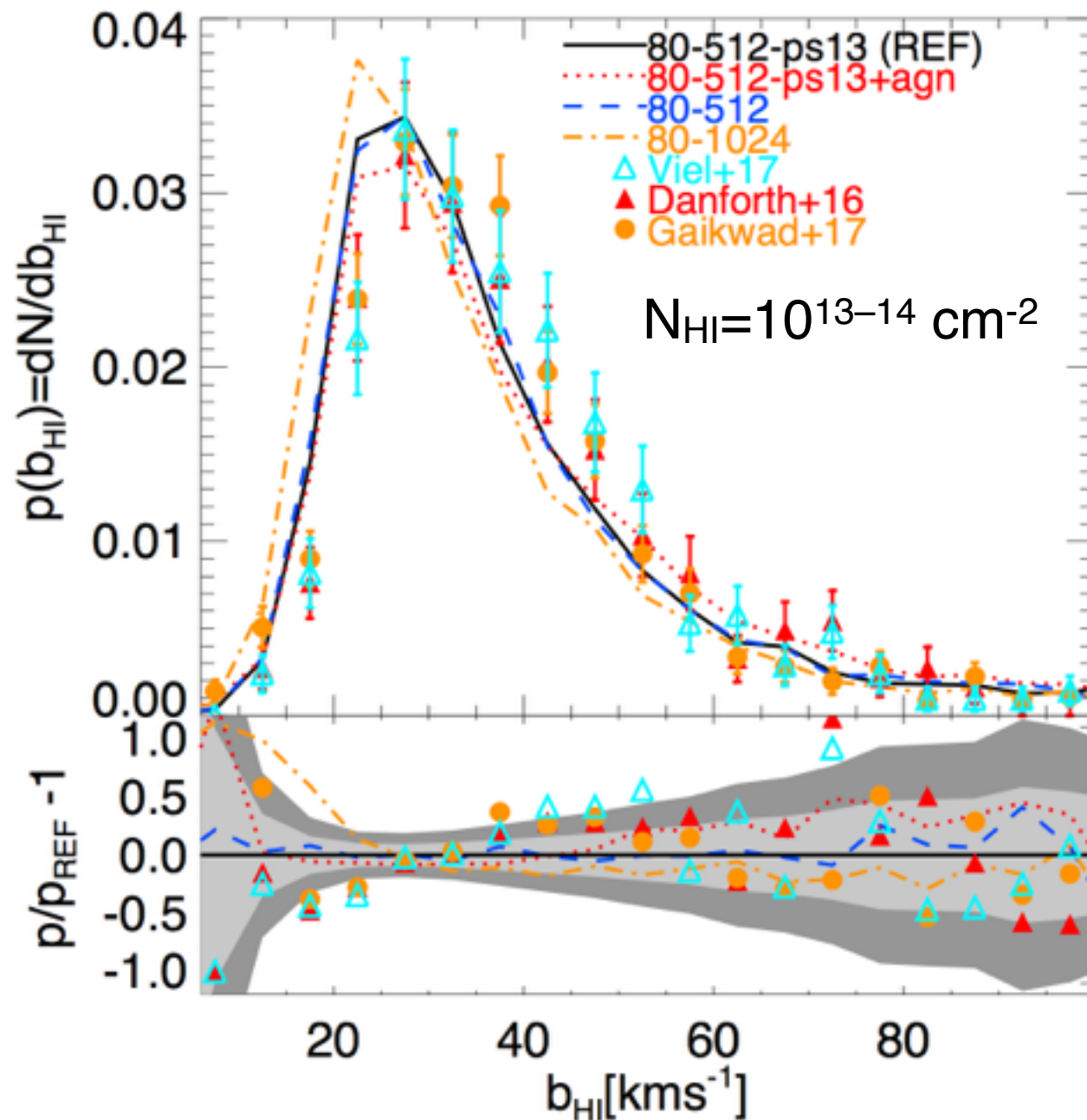
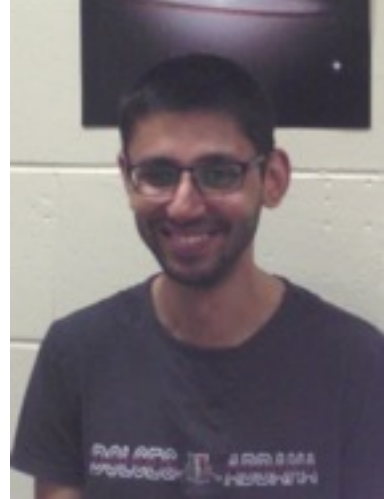
Gas densities



Nasir et al. (2017)

Gas for $N_{\text{HI}} > 10^{14} \text{ cm}^{-2}$ systems has $\delta > 20$ (some may be in CGM)

Line width distribution

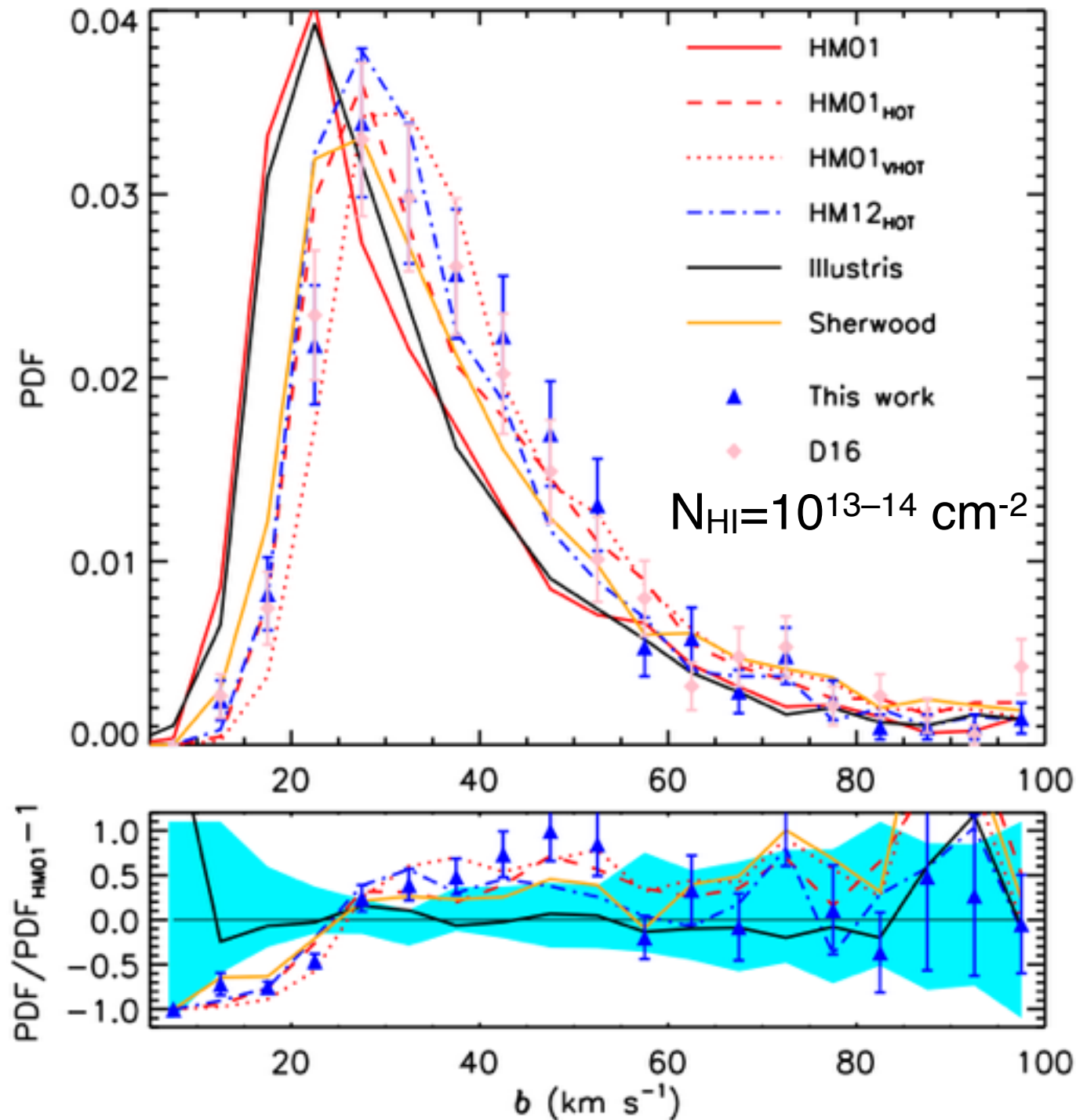


Nasir et al. (2017)

Viel+17 COS data: 704 lines
with $10^{12.5} < N_{\text{HI}} < 10^{14.5} \text{ cm}^{-2}$

- Simulated line widths tend to be too narrow at $z=0.1$, and are strongly dependent on simulation mass resolution;
- AGN feedback produces more lines with $b > 40 \text{ km s}^{-1}$, but not significantly (within expected 1σ variance).

Line width distribution

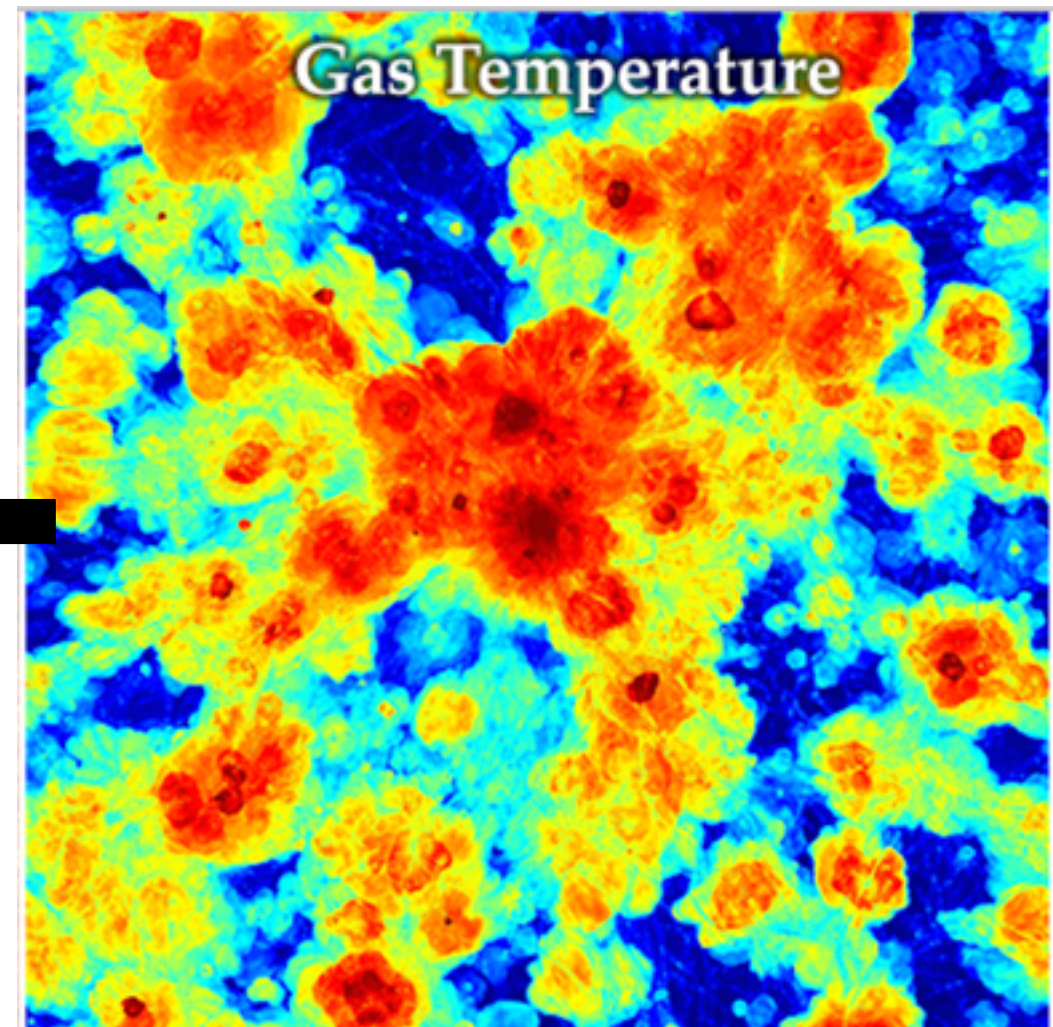
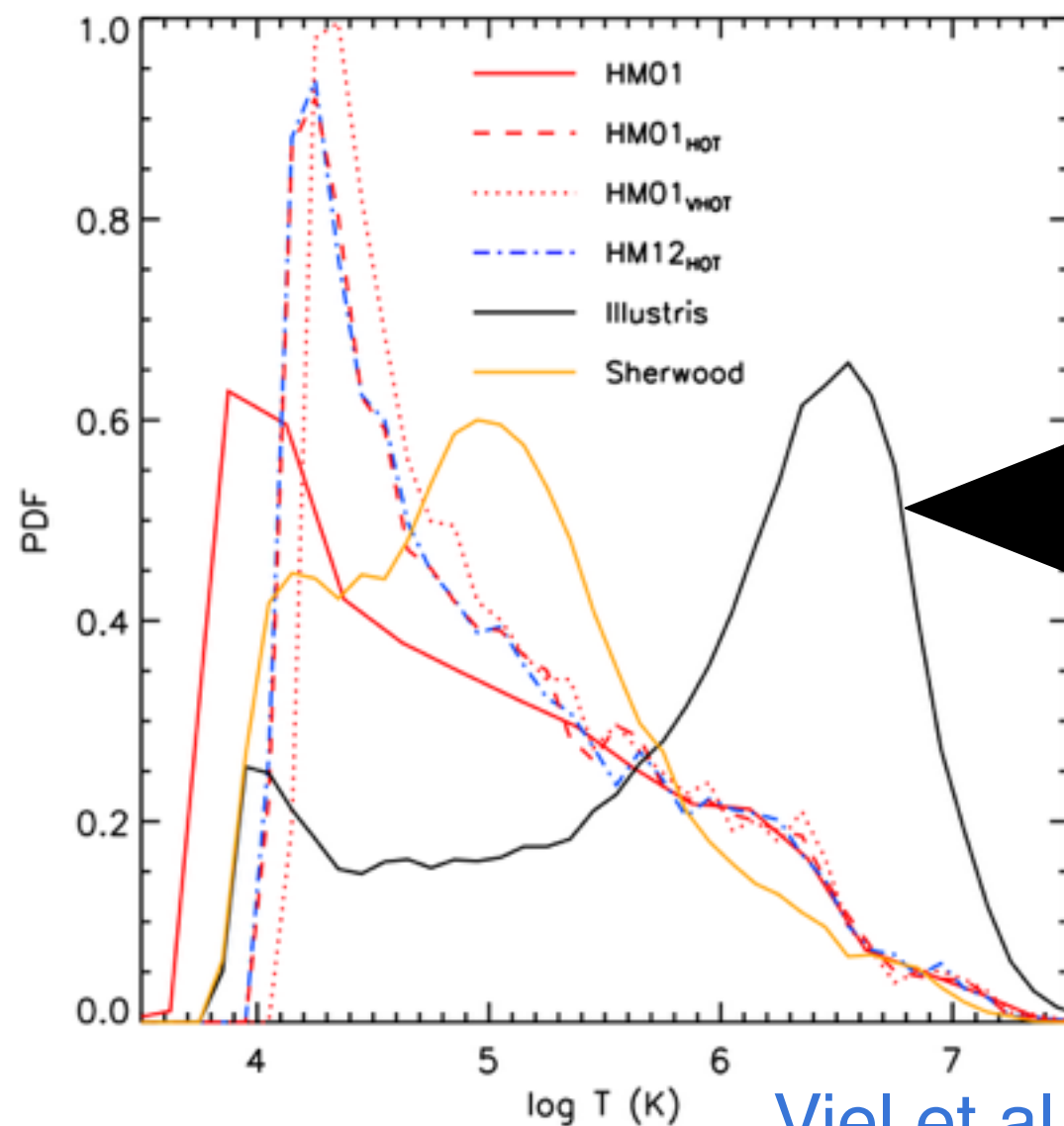


- Illustris line widths also too narrow, despite having very strong AGN feedback and heating (why?);
- Can improve agreement by increasing IGM temperature in models (is this justified?)

Viel et al. (2017)

Gas temperatures

$$N_{\text{HI}}=10^{13-14} \text{ cm}^{-2}$$

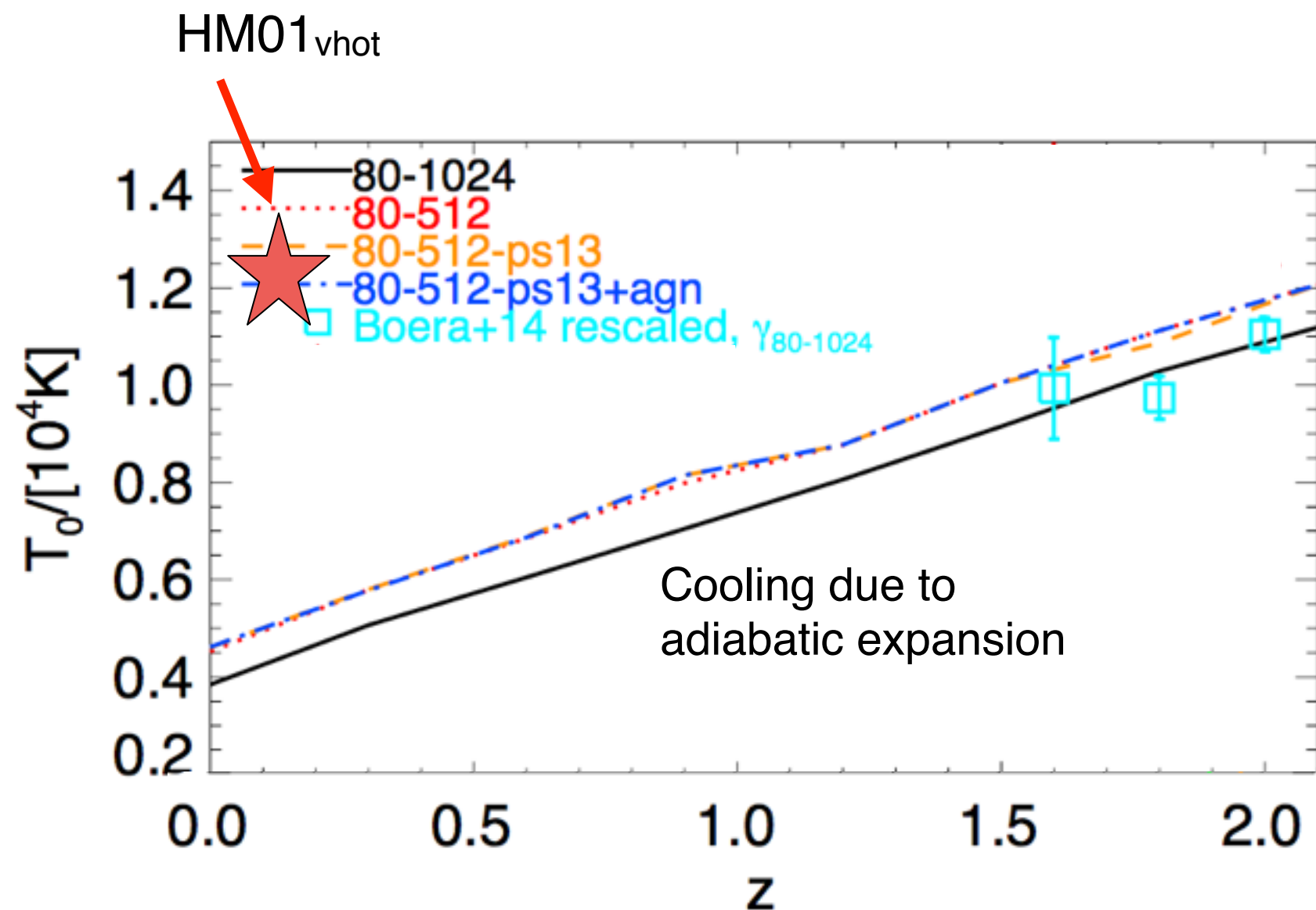
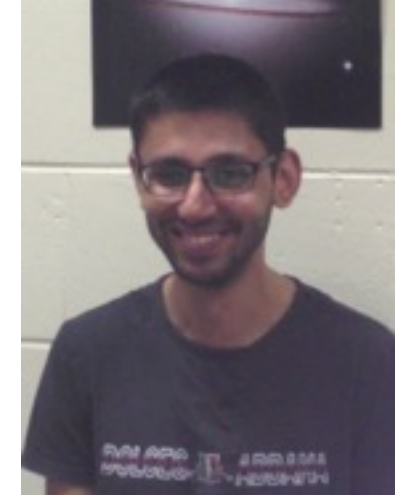


Viel et al. (2017)

Vogelsberger et al. (2014)

Illustris AGN feedback produces a large volume filling factor of $T > 10^6$ K gas at $z \sim 0$ (WHIM). Collisionally ionised $\rightarrow$ no Lyman-alpha absorption.

Temperature evolution



Model	Γ	T_0
HM01	0.127	3.72
HM01 _{hot}	0.127	3.99
HM01 _{vhot}	0.127	4.08
HM12	0.035	3.71
HM12 _{hot}	0.035	3.97
Illustris	0.048	3.73
Sherwood	0.035	3.91

Viel et al. (2017)

Nasir et al. (2017)

Boosting IGM temperature at $z \sim 0.1$ implies additional heating from higher redshift. Inconsistent with thermal history constraints from optical data (can't appeal to He-II heating again, He-II reionisation over by $z \sim 2.5$).

Summary, Part II

Simulations currently struggle to reproduce Lyman-alpha forest absorption in IGM/low density CGM at $z \sim 0.1$. But, **provides useful diagnostic for feedback/IGM models at low redshift.**

Possible solutions?

- Finely tuned feedback that sweeps $\delta > 20$ gas into the IGM at just the right temperature, $T \sim 10^{4.5} - 10^5$ K ?
- An as yet unidentified source of IGM heating at $z < 1.6$ that operates on gas at $\delta > 20$?
- An unresolved (sub-grid) turbulent component to the line velocity widths ([lapichino et al 2013](#), [Gaikwad et al. 2017](#)) ?